

Micro-credentials through the eyes of employers: benefits, challenges and enablers of effectiveness

Nguyen Thi Ngoc Ha, Nina Van Dyke, Michael Spittle, Anthony Watt
and Andrew Smallridge
Victoria University, Melbourne, Australia

Abstract

Purpose – This study explores the perceptions of Australian employers regarding the benefits and challenges of micro-credentials within higher education and enablers of their effectiveness.

Design/methodology/approach – A qualitative approach that included 11 semi-structured interviews with employers affiliated with an Australian university was used. A deductive thematic approach was employed to analyse the data.

Findings – Micro-credentials were generally seen to be beneficial for an array of people, including employees, employers, customers and communities – stakeholders in all environmental layers of micro-credentials' ecological system. Findings also indicated that both challenges of micro-credentials and enablers of their effectiveness depended heavily on attributes of learners, employers and higher education providers. The conclusion is that, based on the evidence of this study, micro-credentials within higher education are worth trialling.

Originality/value – Although research interest in micro-credentials is growing, few empirical studies have investigated micro-credentials' benefits, challenges and enablers of effectiveness, especially from the perspective of employers. The study carries practical and policy implications for those involved with micro-credential research and development.

Keywords Micro-credentials, Ecological systems theory, Higher education, Digital badges, Massive open online courses, Short courses, Professional certificates

Paper type Research paper

Introduction

Micro-credentials are characterised by short study duration, a competency-based approach and small units of learning (Oliver, 2019). They are potentially relevant to learners at all stages of life and supportive of education accessibility and therefore have been advocated as an alternative education pathway (Olcott, 2022; Oliver, 2021a). Simultaneously, micro-credentials have been criticised as a threat to traditional education because they may facilitate gig work and tie education to a particular occupation (Wheelahan and Moodie, 2021a, b). Micro-credentials also lack a globally agreed definition and quality assurance standards (Hijden and Martin, 2023). Following the definition adopted by the Australian National Micro-credentials Framework, this paper defines micro-credentials as certifications of assessed learning that are additional to, alternative to, complementary to, or a component part of, a formal qualification (Department of Education Skills and Employment, 2021).

Micro-credentials are not a new phenomenon, with popular types of micro-credentials, such as digital badges, professional certifications, massive open online courses (MOOCs) and short courses having been successfully implemented for some time (Olcott, 2022). The apparent recent surge in micro-credentials; however, is likely due to governments around

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the world seeking ways to respond to increasing unemployment rates and the significant loss of international tertiary student enrolments caused by the coronavirus disease 2019 (COVID-19) pandemic quarantine measures and travel restrictions (Wheelahan and Moodie, 2021b). Simultaneously, there has been a growing trend towards learning remotely, either synchronously or asynchronously, and increasing demand from employees that learning be provided “just in time when they [employees] need to” (Lee, 2021).

Currently, little research has examined employers’ perspectives regarding the benefits and challenges of micro-credentials within higher education, and enablers of their effectiveness. This study addressed these evidence gaps. Findings from 11 semi-structured interviews with employers across different industries in Australia affiliated with one Australian university offered insights into the potential and practical benefits and challenges of micro-credentials. Participants who had previously completed micro-credential courses themselves also suggested enablers they perceived as central to the effective provision of micro-credentials. Results of this study carry practical and policy implications for those involved with micro-credential research and development.

Literature review

Contemporary work and education contexts

The 21st century workforce and labour market have been associated with the fourth industry revolution, or Industry 4.0 (Szabó-Szentgróti *et al.*, 2021). On the one hand, the spread of computerisation, digitalisation, robotics and autonomous systems has significantly increased the speed, accuracy, power and intensification of work. On the other hand, these factors have reduced labour hours required and replaced many human jobs (Szabó-Szentgróti *et al.*, 2021; Brynjolfsson and McAfee, 2011). This evolution is predicted to create an impact on every employee across a broad range of occupations and professions (Harteis, 2018). Consequently, some skills and occupations may disappear, and others may emerge (Frey and Osborne, 2017). For example, demand for quantitative and statistical skills appears to be decreasing, making room for an expansion of higher cognitive skills such as critical thinking and creativity (Bughin *et al.*, 2018).

Risk in the labour market has largely been transferred from employers to employees (Clarke and Patrickson, 2008). The prevalence of casual, contract and temporary positions indicates the increasing precariousness of work, regardless of career field and complexity level (Bohle, 2016). Jobs will become even more flexible, and future employees will be “flexibly disposable” (Harteis, 2018) due to an expectation of them being capable of a range of tasks while also being permanently contactable. The result is expected to be a shortage of good-quality jobs and intense job competition. Therefore, lifelong learning becomes crucial for enhancing individuals’ human capital and employability (Nimmi *et al.*, 2021).

As human capital theory highlights, learning equals earning. Learning, however, no longer equates to spending time on a tertiary campus listening to lectures (Gauthier, 2020). It has been reported that many higher education graduates fail to work effectively and fall short of employer expectations (Tran, 2018). Consequently, education initiatives such as work-integrated learning and hands-on pedagogy have been implemented, allowing students to study at the workplace and better prepare for future careers (Nguyen and Nguyen, 2022; Nguyen, 2023; Nguyen and Dakich, 2022). The rapid development of communication technology, coupled with repercussions from the pandemic, has accelerated the need to study in informal settings with greater flexibility in time and space (Lee, 2021). In addition, employers often prefer quicker learning approaches to train and prepare their workforce for changing work demands and unpredicted events (Gauthier, 2020). Simultaneously, a skills approach is increasingly popular among job seekers to allow them to quickly reskill, upskill or acquire new skills to remain employable and competitive in the labour market (Burt and Gormley, 2021).

Micro-credentials through the lens of ecological systems theory

Micro-credentials seem to be an appropriate response to the changing nature of education and work by offering flexible, manageable and personalised options to address more focused learning objectives in a short period of time (Prior Filipe *et al.*, 2020). In this study, Bronfenbrenner’s (1979) ecological systems theory was employed to analyse and discuss micro-credentials. The model was conceived to reflect the evolving interaction and accommodation between the developing person and the four environment structures. These are (1) microsystem – the interrelations between the person and environment within the immediate setting; (2) mesosystem – the linkage and interactions between major immediate settings containing the person; (3) exosystem – the process and linkage around the mesosystem influencing the immediate settings; and (4) macrosystem – the broader sociocultural context that covers micro-, meso- and exosystems.

Through the lens of ecological systems theory, micro-credentials can be seen as closely tied to different surrounding environment layers (Figure 1). The recent hype surrounding micro-credentials appears to have been ignited from the macro level, as micro-credentials have attracted a great deal of policy attention. Internationally, UNESCO, the Organisation for Economic Cooperation and Development (OECD), and the European Commission have joined forces to enable micro-credential recognition and portability across nations, including working to bring forward a globally agreed definition of micro-credentials (Oliver, 2019; Wheelahan and Moodie, 2021b). Nationally, micro-credentials have been promoted by many governments, such as New Zealand (New Zealand Qualifications Authority, 2020), Australia (Department of Education Skills and Employment, 2021) and Europe (European Commission, 2020). At exo- and meso-levels, increasing interest in micro-credentials has been generated. For example, in the United States, micro-credentials are strongly supported by early childhood education professionals as providing a clear validation of graduates’ abilities and competencies, especially those who are already employed but seeking contract extension (Gauthier, 2020). However, the connection between micro-credentials and their microsystem seems weak. The research literature indicates low awareness of the term, “micro-credentials”, among employers and students (Nguyen *et al.*, 2023; Gallagher, 2018).

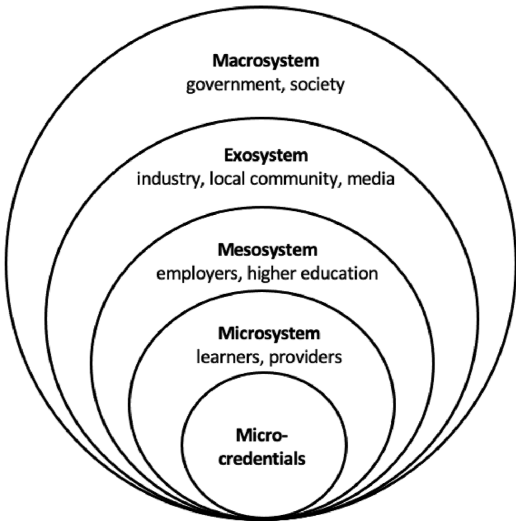


Figure 1.
Micro-credentials
ecological systems

Source(s): Adapted from Bronfenbrenner (1979)

Micro-credentials' benefits, challenges and enablers of their effectiveness

Although research interest in micro-credentials is growing, few empirical studies have investigated micro-credentials' benefits, challenges and enablers of effectiveness, especially from the perspective of employers. This is important because the successful implementation of micro-credentials relies on how key stakeholders including employers perceive micro-credentials (Oliver, 2021b). A recent systematic literature review using the search terms, "micro-credentials", "digital badges", "professional certifications", "MOOCs" and "short courses" resulted in only 29 studies. The review revealed that micro-credentials can bring benefits for learners, higher education institutions and educators (Nguyen *et al.*, 2023). For learners, micro-credentials improve learners' employability capitals (Anderson *et al.*, 2020; Cowley *et al.*, 2021), encourage lifelong learning (Tamoliune *et al.*, 2023), as well as enhance students' study motivation, class engagement and learning performance when they are studied as a component part of a formal qualification (Alt, 2021; Joseph *et al.*, 2021). For higher education providers, micro-credentials are considered an effective channel to strengthen industry cooperation (Dayton *et al.*, 2018), and a prospective vehicle to increase student enrolments and utilise education time and costs (de Moura *et al.*, 2021; Orman *et al.*, 2023). For educators, micro-credentials have been shown to improve their teaching practices and keep them connected with the profession and industry (Deale and Schoffstall, 2015). Likewise, Woods and Woods, 2023 found that micro-credentials designed as subsets of a master's programme bring more benefits than challenges to students, and are also valuable to departments and universities.

Key challenges of micro-credentials lie in the shortage of market demand data, the lack of university leaders' awareness and resources to develop micro-credentials, and the providers' ability to manage a huge number of online learners (McGreal and Olcott, 2022; Ahmat *et al.*, 2021). Previous studies identified additional challenges, such as micro-credentials' content being "superficial and lightweight" (Gilliat-Ray, 2020) and training materials being "confusing" (Cowley *et al.*, 2021). Another challenge to the successful implementation of micro-credentials is the learner's background. While learners with prior knowledge of the topic may complete micro-credentials quickly, novices may find it much more difficult (Peacock *et al.*, 2020). Learners also report time constraints, expensive course fees, and lack of Internet access as inhibitors to micro-credential studies (Rieber, 2017; Shapiro *et al.*, 2017; Deale and Schoffstall, 2015).

Micro-credential effectiveness has been shown to depend on how micro-credentials are developed and delivered, including the extent to which the content is relevant, and the course is enjoyable (Al-Rahmi *et al.*, 2021; Cowley *et al.*, 2021; Tee *et al.*, 2023). Provider reputation (Alraimi *et al.*, 2015; Ashcroft *et al.*, 2021), learner-educator interaction and understanding of the benefits of micro-credentials (Jung and Lee, 2018; Peacock *et al.*, 2020) have proved significant in a learner's decision to undertake a micro-credential. It appears that micro-credentials are more likely to be adopted among educational institutions that are innovative and want to offer more than traditional study programmes (Pirkkalainen *et al.*, 2023). Other factors influencing learners' interest in micro-credentials also include attitudes of other micro-credential learners and social media posts regarding micro-credential usefulness and value (Anderson *et al.*, 2020; Gilliat-Ray, 2020; Virani *et al.*, 2020). Notably, active, self-directed learners seem to utilise micro-credentials the most (Gish-Lieberman *et al.*, 2021).

Methodology

This paper explored the benefits and challenges of micro-credentials within higher education and enablers of their effectiveness from the perspective of 11 Australian employers

affiliated with one Australian university. It reports qualitative findings from a larger mixed-methods study that investigated the viability and effectiveness of micro-credentials. Quantitative results addressing different research questions will be reported in future publications. In this study, we defined “benefits” as advantages gained by undertaking micro-credentials, “challenges” as difficulties inhibiting the successful implementation of micro-credentials, and “effectiveness” as achieving desired learning outcomes and learner satisfaction (Nguyen *et al.*, 2023).

This study had three primary research questions:

- (1) What are the benefits of micro-credentials and for whom?
- (2) What are the challenges of micro-credentials?
- (3) What factors enable the effectiveness of micro-credentials?

Research design and participants

A qualitative approach was used as it is ideal for understanding perceptions and experiences (Patton, 2014). The 11 employers involved in this study were selected purposefully (Creswell and Creswell, 2017). All were industry professionals engaged as members of programme/course advisory groups or alumni of one Australian university. The participants worked in a variety of industries, and 10 of them had previously completed micro-credentials study themselves. As Hennink and Kaiser (2022) found in their systematic review of sample size for saturation in qualitative research, 9–17 interviews reached saturation with studies with homogenous populations. Table 1 provides name codes and details of participants.

Data collection

Data collection was conducted from August to October 2022, after approval through the university’s Human Research Ethics Committee (HRE22-002). Eleven employers expressed willingness to participate in the qualitative component of the study through an online survey (the quantitative component of the larger mixed-methods study), which was sent via email by the university alumni system and relevant university course chairs/directors. Then these employers were contacted and informed via email about the details of the study, ethics information and the consent form. They were invited to participate in online, individual, semi-structured interviews, in which the interviewer prepared and asked questions to elicit responses from participants about their perspectives on micro-credentials (Brown and

Table 1.
Participant details

Participant	Employment field	Sector	Size of organisation	Micro-credentials experience
E1	Education	Public	>100	No
E2	Digital Media	Private	<10	Yes
E3	IT & Computer	Private	10–100	Yes
E4	Education	Public	>100	Yes
E5	Education	Public	>100	Yes
E6	Design	Public	>100	Yes
E7	Public Administration and Safety	Public	>100	Yes
E8	Sport and Exercise	Not-for-profit	<10	Yes
E9	Arts and Humanities	Public	>100	Yes
E10	Business	Public	>100	Yes
E11	Business	Private	10–100	Yes

Source(s): Authors’ own work

Danaher, 2019). The interview schedule was developed after reviewing the research literature and contained a list of open-ended questions that were divided into four sections: participant background, micro-credentials' benefits, micro-credentials' challenges and factors influencing the effectiveness of micro-credentials. Each Zoom interview session lasted approximately 30–40 min and was recorded and transcribed with the consent of participants.

Data analysis

To validate the accuracy of the transcription, a member-checking process (Creswell, 2019) was applied, in which the transcripts were sent back to the participants to ensure the accuracy of the interview contents. In the data analysis stage, a deductive thematic approach was employed (Creswell and Creswell, 2017) using NVivo 12 software. Text segments mentioning benefits, challenges and enablers of micro-credentials' effectiveness were identified, noted with codes, and then grouped into themes (as per research questions) and sub-themes. For example, to answer the first research question, all text segments detailing advantages gained by undertaking micro-credentials were gathered into a theme (or a node in NVivo). Based on the ecological systems theory, they were classified into sub-themes (or child nodes in NVivo) relating to micro-credential stakeholders, such as benefits for employees and employers. The coding process was conducted by the first author, and then themes and sub-themes were discussed with all co-authors. Detailed findings are presented in the next section.

Findings

This section reports the benefits and challenges of micro-credentials, and enablers of their effectiveness. Table 2 summarises themes that emerged from the thematic analysis

Benefits of micro-credentials

Micro-credentials were generally perceived as beneficial for employees, employers, customers and communities. Participants frequently reported updating knowledge and skills as a value of micro-credentials. For graduates without much work experience, micro-credentials were discussed as helping prepare them for workplace skills and requirements, as

Themes	Sub-themes
Benefits of micro-credentials	• Provide knowledge and skills for employees • Support employers' service and revenue • Indirectly benefit customers
Challenges of micro-credentials	• Contribute to community post-pandemic • Low awareness of micro-credentials • Doubt about the value of micro-credentials • Limitations of short study duration • Difficulty in gaining industry recognition • Competitiveness from in-house training, unaffordable price
Enablers of micro-credential effectiveness	• Learners with diverse backgrounds • Relevant, engaging micro-credential content • Hybrid learning mode; convenient timing • Provider reputation and provider strategies to identify target learners • Self-directed, proactive micro-credential learners • Workplace support

Source(s): Authors' own work

Table 2.
Emergед themes

one participant suggested: “There’s a lot of things and tools . . . that upon starting a new role, you could learn from a micro-credential rather than having to try and learn on the job” (E7). Micro-credentials were also considered an effective strategy to support people traditionally disadvantaged in the labour market return to the workforce, especially after the COVID-19 pandemic. Commenting on this, one of the participants highlighted: “[F]or example, women in the workforce who over the last couple of years have probably been impacted greatly [by the pandemic], their knowledge is not current, so bring them up to speed and then help them re-enter the workforce” (E6). For employees, participant E3 suggested: “Given that every year we need to do professional development. I think professional development can be in a micro-credential”. Micro-credentials were also thought valuable for employees considering changing careers, likely because of the potential time and cost efficiency of micro-credentials, as participant E6 confirmed: “I have studied micro-credentials because I have changed careers and worked in seven completely different industries”.

According to all participants, micro-credentials directly support employers’ service and revenue. Participant E2 suggested that micro-credentials seem to be beneficial for both employees and employers: “I think it’s a win-win situation because they [employees] get more knowledge and I get more skills to be implemented in the project”. One participant stated that offering micro-credentials would be an effective way to demonstrate employer care for staff: “If the employer offers that [micro-credentials], I think that would improve the workplace, not because they’re doing the credential, but because the employees feel like the employer cares about helping provide them with professional development” (E4).

In addition, micro-credentials were perceived to indirectly benefit customers. For example, one participant discussed how a micro-credential in Auslan sign language could support early childhood teachers in working with all children: “skills that you do in your micro-credentials don’t just have to be used for individual children with special needs, it can be useful for all children. And what that benefits are that all children get an understanding and awareness” (E3). Furthermore, participant E9 suggested the contribution of micro-credentials to the community, especially after the COVID-19 pandemic: “if we make some [micro-credentials] for free . . .that would be immeasurably, or at least enormously helpful to a community that struggles a lot of ways”.

Challenges of micro-credentials

Challenges of micro-credentials included low awareness of micro-credentials, doubt about the value of micro-credentials, limitations of short study duration, difficulty in gaining industry recognition, competitiveness from in-house training, unaffordable price and learners with diverse backgrounds.

Low awareness of micro-credentials stood out as a key challenge, as participant E7 stated: “There is scope for us to do professional development that includes micro-credentials. But I wouldn’t say it’s integrated into our development plan or anything, mostly because it’s fairly new and there’s not much information about what is possible and what isn’t possible”. There was doubt about the value of micro-credentials, especially when it was hard to find a widely agreed definition: “I heard some people speak about them and I think that confusion is where it’s hard to determine what value they can bring if they don’t really have a definition” (E7). Moreover, some participants felt the short duration of most micro-credentials limits knowledge absorption: “even though it’s like a lot of content but they can’t go in depth into a lot of stuff. It’s like a lot of information comes quickly”. Employers also questioned the value of micro-credentials as stand-alone certifications of competency: “I’m not sure that an employer would accept micro-credential as a real qualification. Like the project management one that I did, but I don’t think any employer would hire me as a project manager because of that” (E4).

As such, effective marketing strategies by training providers may be required, as stated by one participant:

I think it's something that's still evolving, and I think it's still an area that a lot of employers would look at and not quite sure what that is. So, it would benefit from a bit of a marketing campaign around what micro-credentials are. I think if you look in the IT world, they seem to be far more widely accepted then in some of the other areas. (E10)

Participants felt that industry recognition was important, as this was seen as ensuring that micro-credentials aligned with industry requirements and therefore increased the value of micro-credentials. This recognition, however, could be a challenging task for micro-credential providers: "that's the hard part, to get the endorsement, the imprimatur from some of the industry characters" (E8). Moreover, micro-credentials may be less attractive to learners working in companies where in-house training courses are available. Participant E6 confirmed: "we've got a whole academy there that I can just go on anytime to do that. They've got work training and that's like mandatory for all employees". That is probably a reason why, for micro-credential learners, the course's price matters. One participant who had experience running micro-credentials said that the expensive price inhibited learners from enrolling in the courses:

We developed a programme. It was developed to be run at a residential setting outside of Melbourne, so fully catered and it just became too expensive, I think . . . And when it came to promoting the course people didn't sign up and so it didn't run. It was going to cost around \$5000, so it was quite expensive. (E8)

Another challenge facing micro-credentials is the ability to cater to learners from diverse academic backgrounds and work experience. Participant E7 took an example: "I think it's a bit timelier and a bit more relevant to specific things, but it does mean that sometimes they [micro-credentials] are over specific and sometimes too general for me". The possibility of studying and completing a micro-credential probably depended on learners' self-directed study: "I know that [completing a micro-credential] would be difficult for some people, it's not a problem for me. I find it easy to find what I dedicate time to, what I need to do, and then moving forward" (E10).

Enablers of micro-credential effectiveness

The analysis indicated that relevant, engaging micro-credential content; hybrid learning mode; convenient timing; provider reputation and provider strategies to identify target learners; self-directed, proactive micro-credential learners; and workplace support were key enablers of micro-credential effectiveness.

A key factor thought to enable the effectiveness of micro-credentials was the way in which micro-credentials were designed and implemented. One participant suggested that companies' own real data and information could be used as inputs into micro-credential design: "That's quite important to use data and information that are relevant to the persons' day job. If you want to try or you want people from a company to attend, it would be good to use their data as examples, it would be relevant to them" (E11). Another design-related factor discussed by participants as an enabler of effective micro-credentials was high-quality and engaging content, as participant E6 underscored: "I'll pick and choose what is of interest to me. It has to be interesting because otherwise if I sit in front of a video for two hours, I'm going to fall asleep".

Instructions mode was also important, with a blended learning mode recommended by many participants. One participant confirmed: "I think people are willing to engage in so many different types of learning modes these days and so it really needs to look at the blended sector" (E8). Another identified enabler of effective micro-credentials was the convenient timing of micro-credential offerings – for example, by offering micro-credentials

during ebbs in the yearly workflow. One participant provided an example from the sport and recreation industry: “For instance, the football season goes from sort of April to September, and then they have time to do things from October to March. But then cricket is the exact opposite, so it’s important to understand the market accessibility” (E8).

The 10 participants who had studied a micro-credential themselves said provider reputation was important when searching for and deciding where to study a micro-credential. These employers/learners preferred micro-credentials offered by recognised training organisations: “I don’t think I would engage in something that from a place that I didn’t know . . . not necessarily the best but should be recognisable” (E7). Some participants also underscored that the successful implementation of micro-credentials relies on the ability of training providers to identify micro-credential learning demands and understand the needs of their target learners. For example, participant E10 said: “They [micro-credential providers] got to really target to this audience, with this product”. Another participant added: “[Y]ou may even need to do some really effective market research to determine whether there is a demand” (E8).

Regarding micro-credential learners, self-directed learning may be a success factor. This was confirmed by participant E10: “I do a lot of study and training. I don’t find it [a micro-credential] particularly much of a challenge for me. I find it easy to find that I’m dedicating time to do what I need to do, and then moving forward”. Participants also noted that micro-credential students are usually lifelong learners who are open-minded and proactive regarding self-improvement. One participant commented: “Because micro-credentials are building knowledge. And then if you have the right mindset, the right objectives, and the right focus on micro-credentials, you can implement them and do good things with them. If you don’t, you cannot do much with them” (E2).

Workplace support was identified as an important enabler to the successful implementation of micro-credentials. As one participant suggested, when the whole team attended a micro-credential course, they needed workplace support in terms of time and work allocation. Participant E11 commented that “because it was the whole team attending, we needed to make sure we add a block of three days that we could get someone else to stand in for us at work”. Two participants confirmed that funding was available at their workplace to sponsor employees studying micro-credentials: “I can support them in terms of paying for the course” (E2) or “I sent people on leadership courses” (E11).

Discussion

This study used a qualitative methodology to explore the perspective of Australian employers affiliated with one Australian university regarding the benefits and challenges of micro-credentials, and enablers of their effectiveness through the lens of the ecological systems theory model (Bronfenbrenner, 1979, 2005). Results from the 11 semi-structured interviews revealed that micro-credentials were generally seen to be beneficial for an array of people, including employees, employers, customers and communities – stakeholders in all environmental layers of micro-credentials’ ecological system. It is also worth noting that half of the participating Australian employers strongly confirmed the actual use of micro-credentials in their career progression and their sponsorship for employees who undertook micro-credentials as part of professional development. However, there are challenges to the wide-scale adoption of micro-credentials and findings revealed that their effectiveness was dependent on attributes of learners, employers and higher education providers.

Micro-credentials: worth a try?

Micro-credentials are not without challenges. This study revealed that employers’ low awareness of micro-credentials and doubt over their values stood out as key barriers

inhibiting the successful implementation of micro-credentials. Actions are required to “ensure they [micro-credentials] are valued, and bring value to key stakeholders, particularly learners, employers and providers” (Oliver, 2021b). For this to happen, Varadarajan *et al.* (2023) called for cooperation and interaction among learners, employers, higher education providers and the government to mitigate challenges in the implementation of micro-credentials.

More significantly, this paper reinforced that micro-credentials are not for everyone, although they are more accessible and manageable as compared to traditional tertiary education degrees. Consistent with previous studies (Deale and Schoffstall, 2015; Shapiro *et al.*, 2017; Peacock *et al.*, 2020), this research found that designing micro-credentials for potential learners with diverse academic backgrounds is challenging. Findings indicated micro-credential barriers associated with learners including learners with no or little prior knowledge of the content, difficulties absorbing content within a short course duration, and preferring in-house training to micro-credentials.

This study, however, recommends that micro-credentials within higher education are worth pursuing. Previous studies (Nguyen *et al.*, 2023; Woods and Woods, 2023) suggested that micro-credentials are valuable to university stakeholders, including students, educators and departments. This research partly reflected these results by confirming benefits to learners associated with career exploration and progression (the microsystem, according to ecological systems theory). The potential advantages of micro-credentials, including short study duration, small and focused units of learning, and a competency-based learning approach (Olcott Jr, 2022), allow learners at different stages of life to quickly reskill, upskill or acquire new skills. Unlike macro-credentials (e.g. bachelor’s and master’s degrees), micro-credentials are not often listed as compulsory qualification requirements in job selection criteria (Kwok *et al.*, 2012; Oginni *et al.*, 2022). However, they can validate employees’ current abilities and skills (Gauthier, 2020), are better indicators of specialised skills and knowledge, and focus more on job issues than do university degrees (Gallagher, 2018).

The findings from this study further indicated that micro-credentials were seen as directly supporting employers’ services and revenues (mesosystem) as well as benefiting customers (exosystem). This study also highlighted that offering micro-credentials may be a practical way to exhibit employer care for staff and government support for the community (macrosystem), especially for individuals who are impacted by unexpected events like the COVID-19 pandemic. These findings are contrary to the argument of Moodie and Wheelahan (2020), who doubted the capability of micro-credentials to support learners’ careers, citizenship and participation in society. In line with the ideas of Oliver (2019), this paper highlighted micro-credentials’ contribution to social inclusion and education accessibility and suggested the potential of micro-credentials as a channel for further involvement of industry and government in tertiary education and employees’ well-being.

Micro-credential enablers: key role of microsystem stakeholders

This study confirmed that the effectiveness of micro-credentials in higher education relies largely on learners and training providers – stakeholders in the ecological microsystem. As compared with traditional tertiary courses, micro-credentials, especially those implemented on online platforms, may cater for more flexibility in that they allow learners to study anywhere, anytime, at their convenience (Martinez-Marroquin and Male, 2021). Many micro-credential learners have multiple priorities in their lives (Nguyen *et al.*, 2023) and diverse academic backgrounds (Phan *et al.*, 2016; Shapiro *et al.*, 2017). Previous studies (Rieber, 2017; Hone and El Said, 2016) revealed that only a small percentage of micro-credential learners show commitment to complete, and learners tend to drop out before or at the middle point of

the course. The findings of our study corroborate those of [Gish-Lieberman et al. \(2021\)](#), who underscored that learners who are self-directed, autonomous and active in learning seem to complete and utilise micro-credentials the most.

This study also supports prior evidence that the reputation of higher education providers is a factor attracting learners' interest and intention to study micro-credentials ([Alraimi et al., 2015](#); [Cowley et al., 2021](#)). This research found that additional education provider-related factors influencing the successful implementation of micro-credentials included marketing and pricing strategies to target learners as well as initiatives to gain industry recognition. Confirming prior research, learners' micro-credential retention and completion rates also depend on the quality of the micro-credential design and implementation ([Al-Rahmi et al., 2021](#), [Albelbisi and Yusop, 2019](#)). Our study identified four key design and implementation components: (1) high-quality, engaging and relevant content, (2) blended learning mode, (3) appropriate timing and (4) affordable price. Higher education institutions must address new challenges arising from the distinctive, flexible nature of micro-credentials and the unpopularity of the term among potential learners and employers ([Gallagher, 2018](#); [Ashcroft et al., 2021](#)).

Employer support: a catalyst for micro-credentials' effectiveness

Another key finding from this study was evidence that employer support is viewed as a catalyst for the successful development and undertaking of micro-credentials in higher education. Employer voice has proved vital in education strategies aimed at enhancing graduate employability, such as work-integrated learning ([Nguyen and Dakich, 2022](#); [Nguyen, 2022](#)), and its importance also extends to micro-credentials. According to [Gallagher \(2018\)](#), micro-credentials reflect the employees' commitment to lifelong learning and skills development. Therefore, it is crucial that employers recognise and support employees' endeavours to study micro-credentials as part of their professional development and career progression. As reported in this study, funding, time and work allocation are needed, especially when the whole team attends a micro-credential course.

In addition, this study found that industry recognition and involvement with training providers in the entire development process of micro-credentials were perceived to be of crucial importance to their effectiveness because learners value industry-aligned courses with quality verified by employers ([Gallagher, 2018](#)). It has been argued that the involvement of employers in the development of micro-credentials is not just a choice but their responsibility, especially in the context that "employers needing to fill roles more quickly are unable to wait for certain skills to evolve their way into the workforce" ([Burt and Gormley, 2021](#)), and skill-based hiring approaches are increasingly popular ([Gallagher, 2018](#)). As a stakeholder in the meso-/exo-layers of micro-credentials' ecological systems, the employer/industry voice represents the influence of the translation of micro-credential policy (macrosystem) into practice (microsystem).

Limitations, strengths and implications for future research

Limitations of this study included the small sample size – 11 employers, and that all were affiliated with a single Australian university. Strengths included the diversity of industry fields represented and that 10 of the 11 participants had themselves studied a micro-credential and thus were able to provide both employer and learner perspectives. Future research should include a larger and broader array of voices – such as those of students and other university stakeholders, including university leaders, lecturers and professional staff – to provide additional insights into the practices, benefits, challenges and enablers of the effectiveness of micro-credentials. Methodologically, quantitative work needs to be done to explore the extent micro-credentials were adopted and accepted, especially among learners

and employers. Longitudinal studies are welcome to evaluate the long-term effects of micro-credentials in higher education.

Conclusion

This study explored the benefits, challenges of micro-credentials and enablers of their effectiveness. It is one among few studies that investigated micro-credentials from the perspectives of Australian employers and suggested that micro-credentials within higher education are worth trialling. Specifically, this study found that employers perceived micro-credentials as having the potential to support employees in meeting professional development requirements, preparing for the future workplace, returning to the workforce and changing careers, while also supporting employers in providing services and revenues, benefiting customers and contributing to community engagement. The effectiveness of micro-credentials in achieving these positive outcomes for employees and employers was identified as reliant on factors such as micro-credential design and implementation, quality of the training providers, attributes of learners and workplace support. Micro-credential developers and related stakeholders should take these enablers into consideration when designing and implementing micro-credentials. To maximise their potential, the provision of “micro-credentials” needs to be popularised. In practice, the involvement of industry and professional associations with higher education institutions in developing and certifying micro-credentials would be a decisive step to make this happen.

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Corresponding author

Nguyen Thi Ngoc Ha can be contacted at: ha.nguyen@vu.edu.au

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